

Data Modeling: CSCI E-106

Harvard Extension School – Spring 2025

Lecture # 1

What is a Data Scientist?

What is a Quant? A quant is someone specializing in the implementation of statistical and mathematical methods to understand and forecast the financial markets' behavior.

Traditionally, quants have a background in the following fields

- Statistics, Economics, Mathematics, Physics and Finance.

With the recent evaluation of Machine Learning Models, Large Language Models (LLM) such as ChatGPT and Copilot. New skill sets are needed for these types of models.

You

what is data scientist?

Copilot

A **data scientist** is an analytical data expert who uses data science to discover insights from massive amounts of structured and unstructured data. They play a crucial role in shaping or meeting specific business needs and goals by extracting meaningful knowledge from complex datasets^[1]. Here are some key aspects of their work:



1. **Understanding Data:** Data scientists work with various types of data, including numerical, textual, and visual information. They use their expertise in statistics, mathematics, and computer science to explore patterns and trends.
2. **Problem Solving:** They apply analytical skills and domain knowledge to solve real-world problems. By asking the right questions and developing hypotheses, they uncover valuable insights.
3. **Predictive Modeling:** Data scientists create algorithms and data models to forecast outcomes. These models can help organizations make informed decisions based on data-driven predictions.
4. **Machine Learning:** They leverage machine learning techniques to improve data quality or enhance product offerings. This involves using tools like Python, R, SAS, or SQL for data analysis.
5. **Communication:** Data scientists communicate their findings and recommendations to other teams and senior staff. Effective communication is essential for translating complex data insights into actionable strategies.

In the United States, data scientists earn an average salary of \$108,659, and the demand for their expertise is expected to grow significantly in the coming years^[2]. If you're interested in this field, consider exploring online courses and certificates to build your skills and embark on a rewarding career! 😊

Learn more [1 cio.com](#) [2 coursera.org](#) [3 en.wikipedia.org](#) [4 careerexplorer.com](#) [+1 more](#)

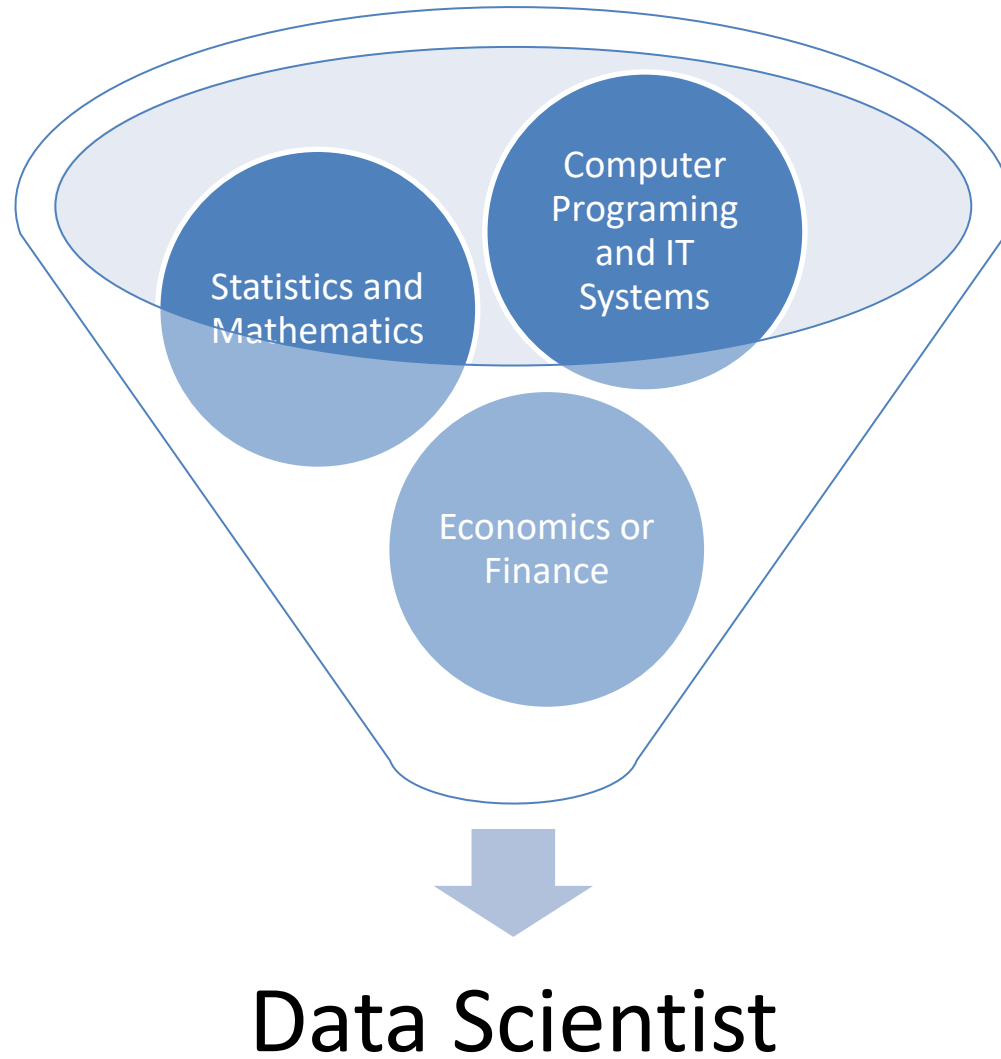
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What skills are essential for a data scientist?

Can you recommend any online courses for aspiring data scientists?

How does a data scientist differ from a machine learning engineer?

What is a Data Scientist?



Introductions

You need to define the problem or understand the problem that you are solving. To be able to do that, you need to do the followings:

- Have an understanding of the subject. If it is financial model you are building, you should develop a basic understanding of finance.
- Understand the objective and identify the limitations. Again, often you will be working with a collaborator who may not be clear about what the objectives are. Beware of “fishing expeditions” — if you look hard enough, you will almost always find something, but that something may just be a coincidence.
- Make sure you know what the client wants. You can often do quite different analyses on the same dataset. Sometimes data scientist perform an analysis far more complicated than the client really needed. You may need to find balance between complexity and transparency.
- Understand your data.
 - How many observations are there? How many variables are there? Time Frame of the data? Is the data complete?
 - Is the data reliable ? Are there any personal identification information (SSN, Name)?
 - Are there missing values? This is a common problem that is troublesome and time consuming to handle.
 - How are the data coded? In particular, how are the categorical variables represented?
 - What are the units of measurement?
 - Are there any outliers? Beware of data entry errors and other corruption of the data. This problem is all too common — almost a certainty in any real dataset of at least moderate size.

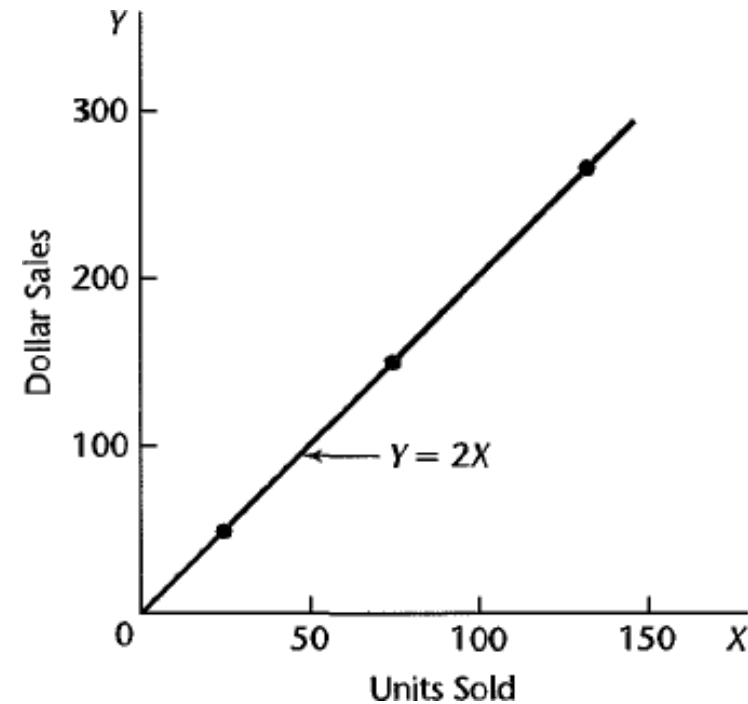
Relation Between Variables

- A functional relation between two variables is expressed by a mathematical formula. If X denotes the independent variable and Y the dependent variable, a functional relation is $Y=f(X)$.
- Given a particular value of X , the function f indicates the corresponding value of Y .

Example 1

- Consider the relation between dollar sales (Y) of a product sold at a fixed price and number of units sold (X). If the selling price is \$2 per unit, the relation is expressed by the equation: $Y=2X$.
- Data is below

Period	Number of Units Sold	Dollar Sales
1	75	\$150
2	25	50
3	130	260



Getting Started with Regression

Regression analysis is the starting point in data science. This is because regression models represent the most well-understood models in numerical simulation. Once we experience the workings of regression models, we will be able to understand all other machine learning algorithms. Regression models are easily interpretable as they are based on solid mathematical bases (such as matrix algebra for example). We will see in the following sections that linear regression allows us to derive a mathematical formula representative of the corresponding model. Perhaps this is why such techniques are extremely easy to understand.

Regression analysis is a statistical process done to study the relationship between a set of independent variables (explanatory variables) and the dependent variable (response variable). Through this technique, it will be possible to understand how the value of the response variable changes when the explanatory variable is varied.

Consider some data that is collected about a group of students, on: number of study hours per day, attendance at school, and scores on the final exam obtained. Through regression techniques, we can quantify the average increase in the final exam score when we add one more hour of study. Lower attendance in school (decreasing the student's experience) lowers the scores in the final exam.

A regression analysis can have two objectives:

Explanatory analysis: To understand and weigh the effects of the independent variable on the dependent variable according to a particular theoretical model

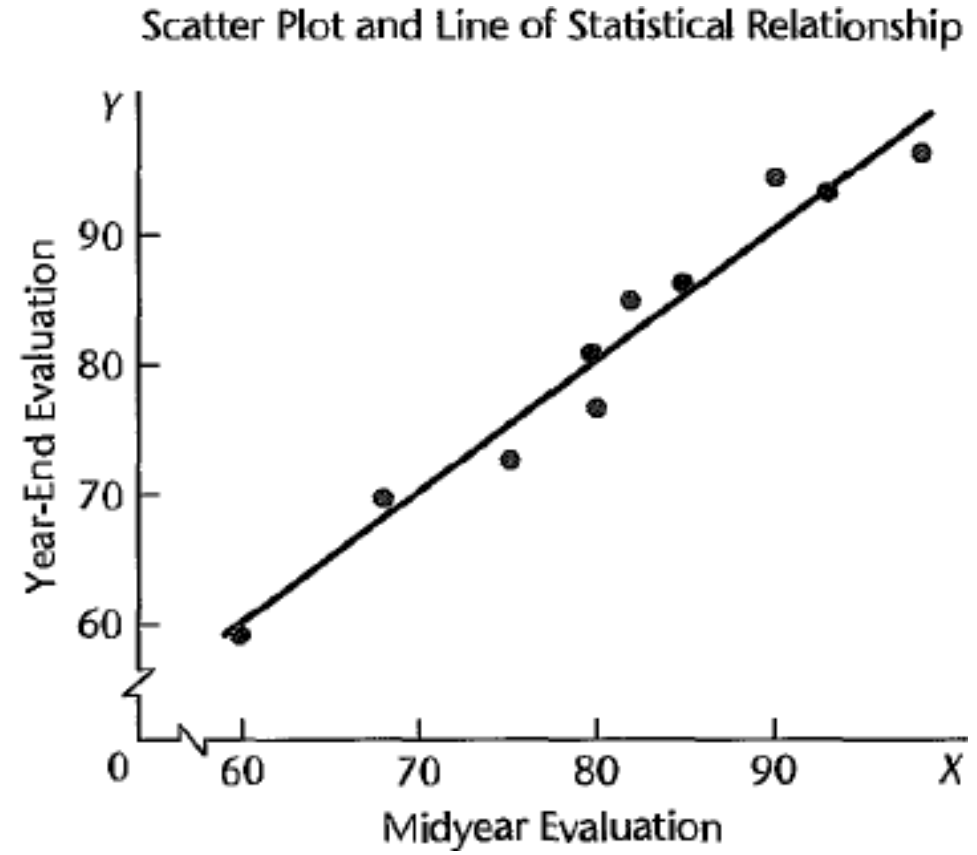
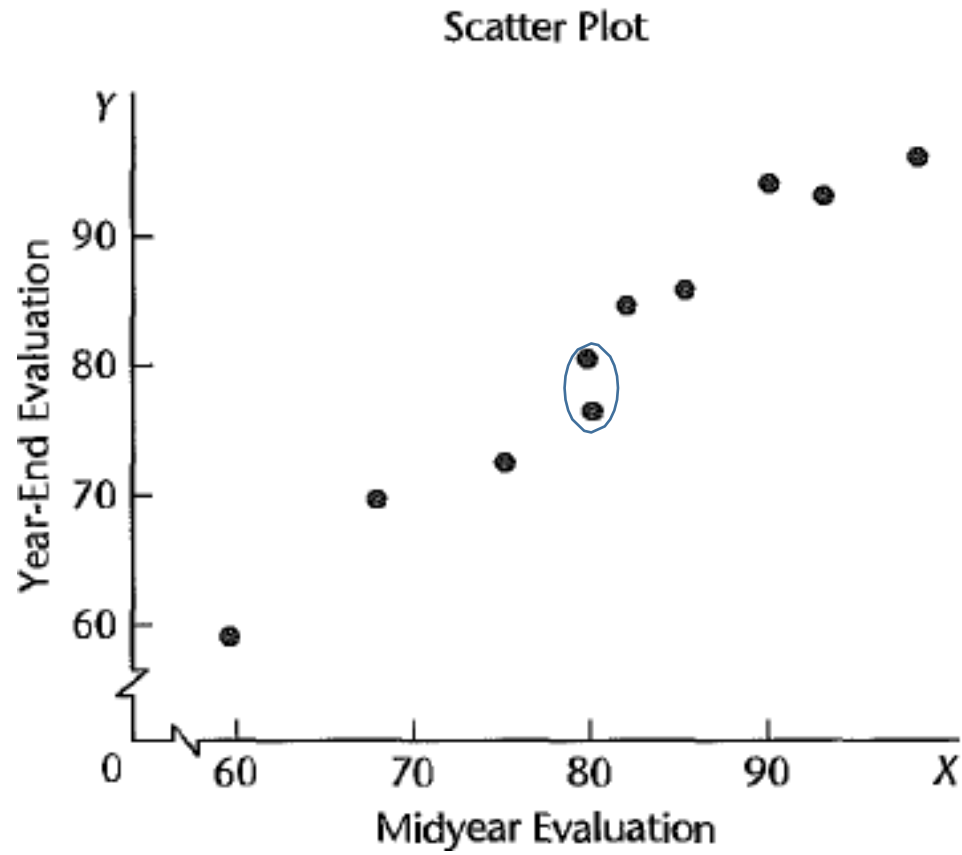
Predictive analysis: To locate a linear combination of the independent variable to predict the value assumed by the dependent variable optimally

Statistical Relation between Two Variables

- A statistical relation, unlike a functional relation, is not a perfect one. In general, the observations for a statistical relation do not fall directly on-the curve of relationship.
- Example 2: Performance evaluations for 10 employees were obtained at midyear and at year-end. These data are plotted in Figure 1.2a. Year-end evaluations are taken as the dependent or response variable Y, and midyear evaluations as the independent, explanatory, or predictor variable X.

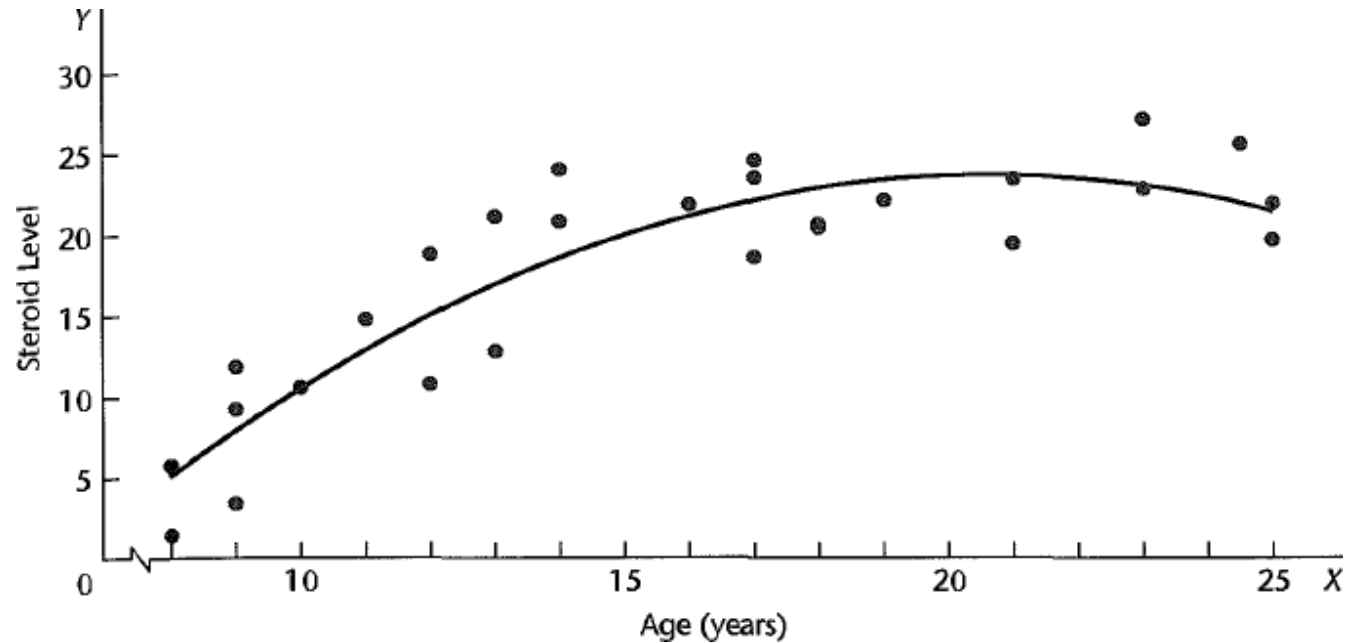
Example 2, cont'd

- Which kind of pattern can be observed in the figure?
- What kind of the relation between midyear and year-end evaluations?



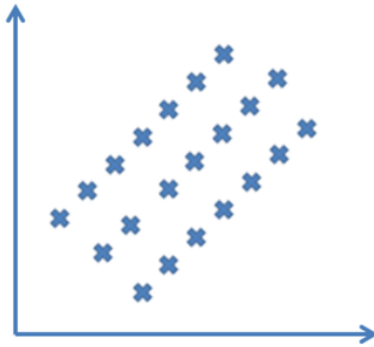
Example 3

- Data, on age and level of a steroid in plasma for 27 healthy females between 8 and 25 years old, strongly suggest that the statistical relationship is curvilinear (not linear).

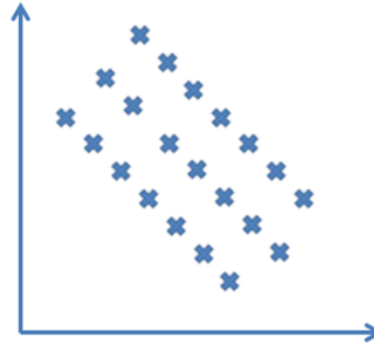


- The curve of relationship implies that, as age increases, steroid level increases up to a point and then begins to level off.
- Note again the scattering of points around the curve of statistical relationship, typical of all statistical relations.

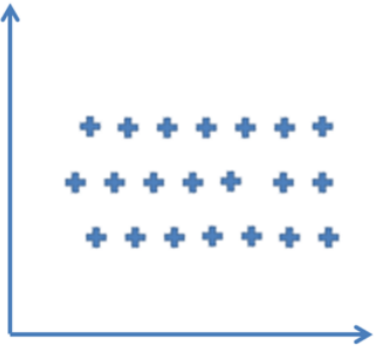
Association between variables – covariance and correlation



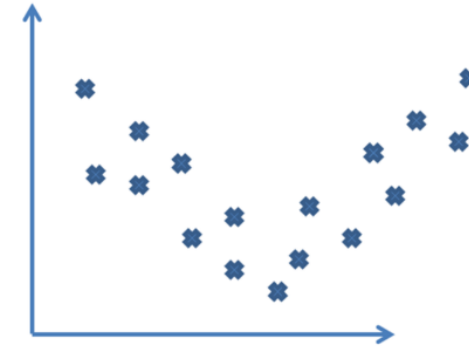
concordant association



discordant association



absence of association



non-linear association

The most known variability index is variance; it measures how far a set of numbers are spread out from their mean. It represents the mean of the squares of deviations of individual values from their arithmetic mean. The Variance is calculated using the following formula:

$$Variance = \frac{1}{N-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{N-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})$$

To measure the association between variable x and variable y we can use covariance. Covariance provides an unnormalized measure of their tendency to move together, and is estimated by summing the product deviations from the mean for each variable in each period. Covariance is calculated using the following formula:

$$cov(x, y) = cov(y, x) = \frac{1}{N-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

The covariance sign indicates the type of relationship between the two variables. A positive sign indicates that they move in the same direction while a negative indicates that they move in opposite directions. Also, the tighter the relationship between the variables, the greater the covariance.

The most known variability index is variance; it measures how far a set of numbers are spread out from their mean. It represents the mean of the squares of deviations of individual values from their arithmetic mean. The Variance is calculated using the following formula:

$$corr(x, y) = \frac{cov(x, y)}{sd(x) * sd(y)}$$

Here:

- $cov(x, y)$ is the covariance
- $sd(x)$ is the standard deviation of x
- $sd(y)$ is the standard deviation of y

The correlation coefficients range between -1 and 1, where:

- Values that are derived closer to 1 are an indication of a positive linear relationship between the data columns
- Values closer to -1 are an indication of one column of data with a negative linear relationship to another column of data (anti-correlation)
- Values that are close to or equal to 0 indicate that a linear relationship between the data columns does not exist

What is Regression?

A regression model is a formal means of expressing the two essential ingredients of a statistical relation:

1. A tendency of the response variable Y to vary with the predictor variable X in a systematic fashion.
2. A scattering of points around the curve of statistical relationship.

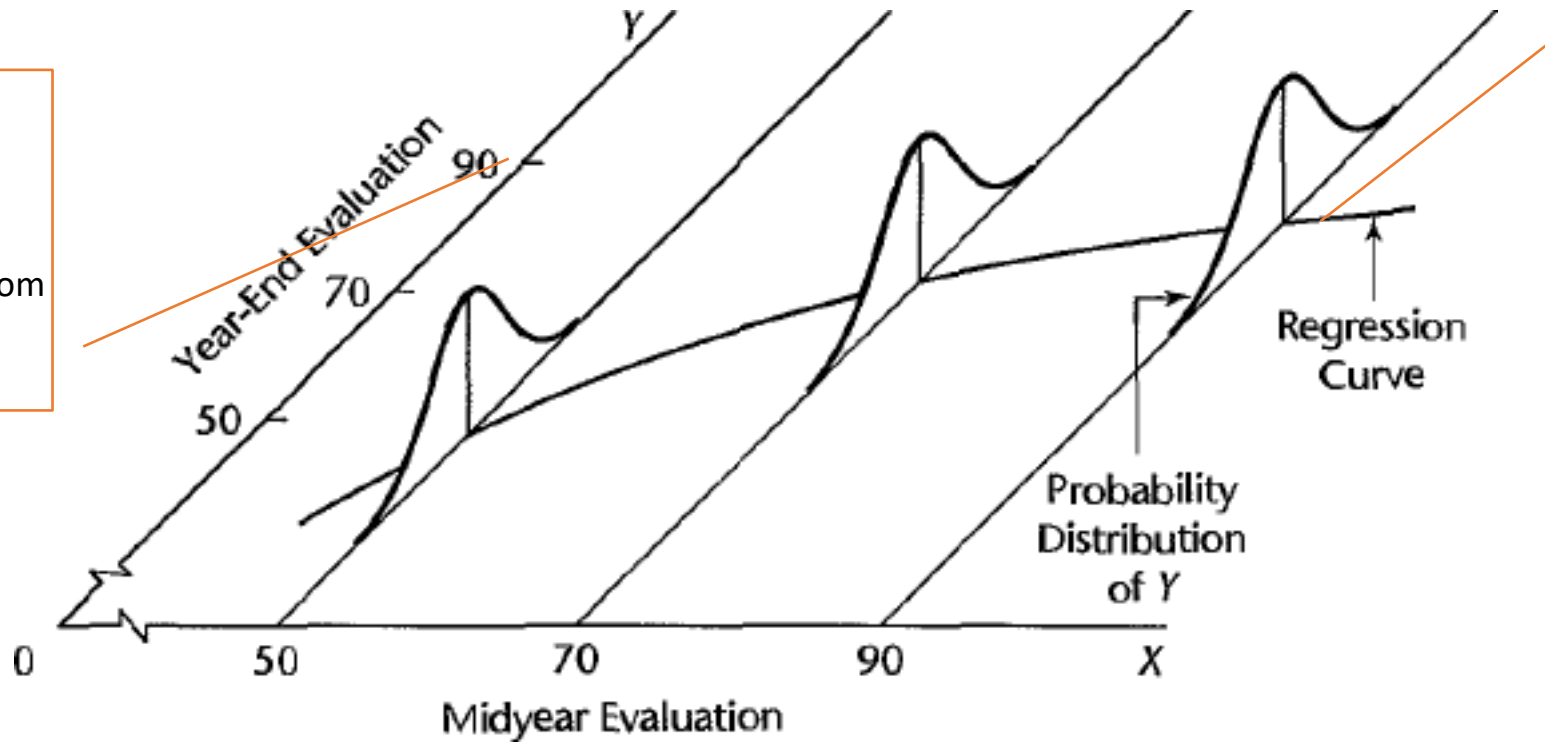
These two characteristics are embodied in a regression model by postulating that:

1. There is a probability distribution of Y for each level of X .
2. The means of these probability distributions vary in some systematic fashion with X .

Example 2, cont'd

- Consider again the performance evaluation example. The year-end evaluation Y is treated in a regression model as a random variable. For each level of midyear performance evaluation, there is postulated a probability distribution of Y .

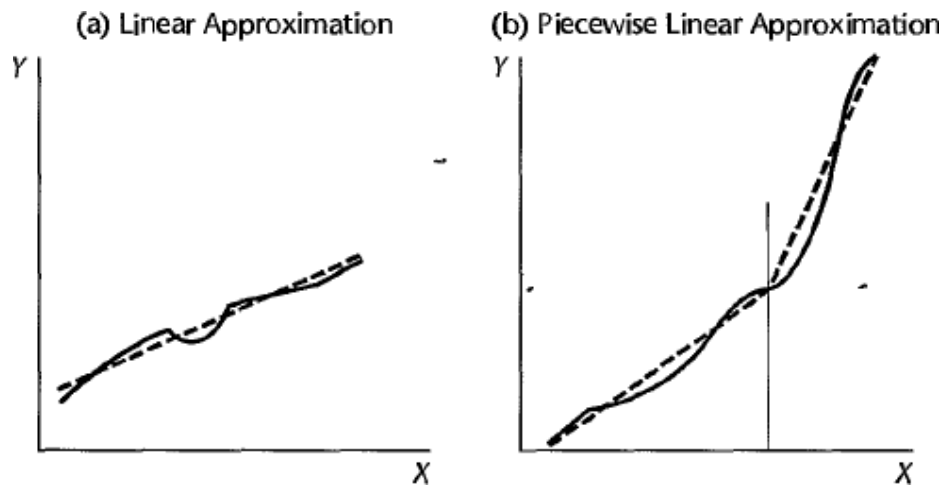
The actual year-end evaluation of this employee, $Y = 94$, is then viewed as a random selection from this probability distribution.



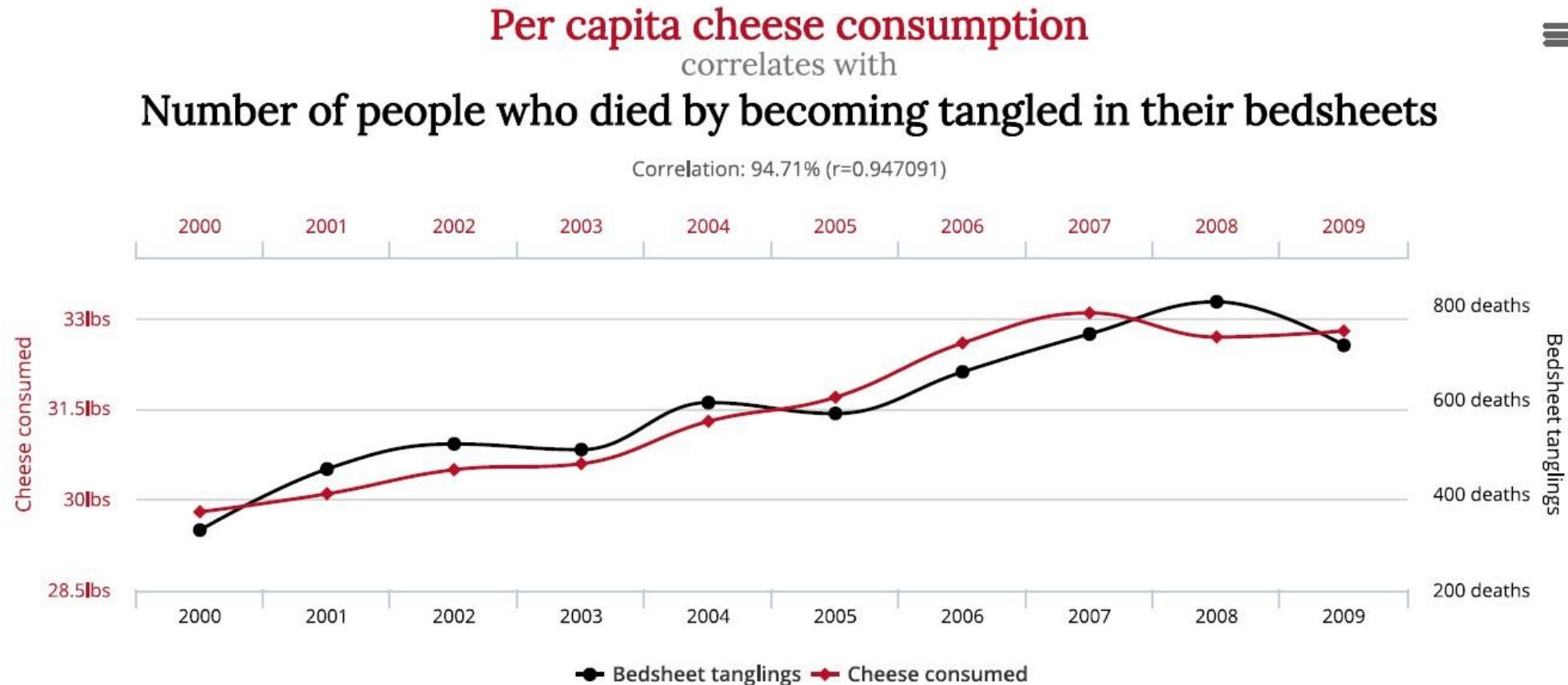
shows such a probability distribution for $X = 90$, which is the midyear evaluation for the first employee

Construction of Regression Models

- Definition of a Dependent Variable or Problem: Regression models could be used for different purposes
 - Analyzing data
 - Prediction
 - Quality Controls
 - Experimental Designs, and etc....
- Data Availability and Data Quality: What are independent variables? And are they accurate?
- Functional relationships linear vs. non linear?



Regression and Causality



tylervigen.com

Data sources: U.S. Department of Agriculture and Centers for Disease Control & Prevention

Regression Model Definition

The linear regression function with one predictor variable:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, \dots, n$$

- Y_i : the value of response variable in the i th trial
- β_0, β_1 : parameters
- X_i : a known constant; the value of the predictor variable in the i th trial
- ε_i : random error term; $E\{\varepsilon_i\} = 0$; $\sigma^2(\varepsilon_i) = \sigma^2$; uncorrelated ($\sigma\{\varepsilon_i, \varepsilon_j\} = 0, i \neq j$)

Features

$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ is the sum of two components:

1. $\beta_0 + \beta_1 X_i$ ----- $>$ *Constant Term*
2. ε_i ----- $>$ *Random Term*

$$E\{\varepsilon_i\} = 0:$$

$$E\{Y_i\} = E\{\beta_0 + \beta_1 X_i + \varepsilon_i\} = \beta_0 + \beta_1 X_i$$

- The regression function:

$$E\{Y\} = E\{\beta_0 + \beta_1 X + \varepsilon\} = \beta_0 + \beta_1 X$$

- The regression function relates the means of the probability distribution of Y for given X to the level of X

Features, Cont'd

- Y_i in the i^{th} trial exceeds or falls short of the value of the regression function by the error term amount ε_i
- The error terms are assumed to be **uncorrelated**, so are the responses Y_i and Y_j .
 - $\sigma^2\{\varepsilon_i\} = \sigma^2$ and $(\sigma\{\varepsilon_i, \varepsilon_j\} = 0, i \neq j)$
 - $\sigma^2\{Y_i\} = \sigma^2(\beta_0 + \beta_1 X_i + \varepsilon_i)$
$$= \sigma^2(\beta_0 + \beta_1 X_i) + \sigma^2(\varepsilon_i)$$
$$= \sigma^2\{\varepsilon_i\} = \sigma^2$$

Example

- A consultant for an electrical distributor is studying the relationship between the number of bids requested by construction contractors for basic lighting equipment during a week and the time required to prepare the bids. Suppose that regression model is applicable and is as follows:

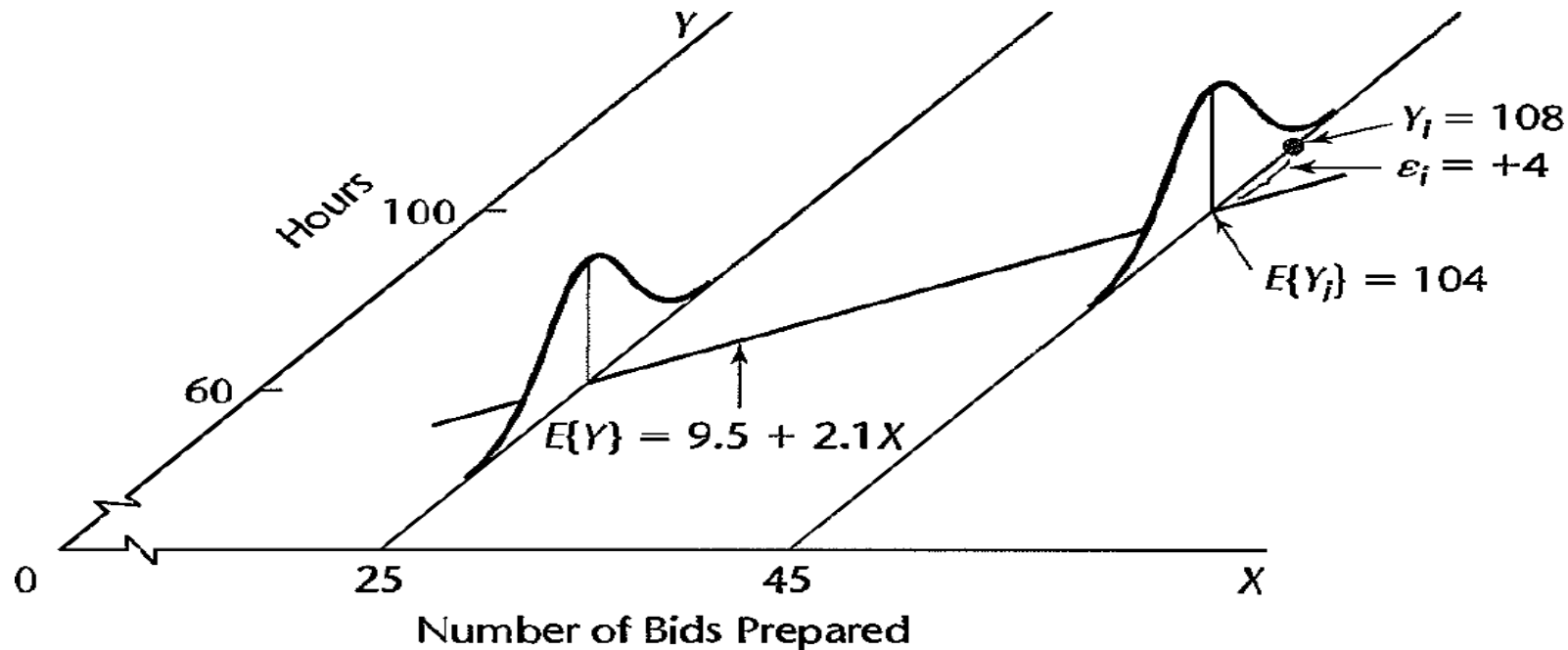
$$Y_i = 9.5 + 2.1X_i + \varepsilon_i \quad \text{and} \quad E\{Y\} = 9.5 + 2.1X$$

- Y is the number of hours required to prepare the bids
 - X is the number of bids prepared in a week
-
- Suppose that in the i^{th} week, $X_i = 45$ bids are prepared and the actual number of hours required is $Y_i = 108$. In that case, the error term value is $\varepsilon_i = 4$:

$$E\{Y_i\} = 9.5 + 2.1(45) = 104 \rightarrow \varepsilon_i = \text{Actual Value} - \text{Predicted Value} = 108 - 104 = 4$$

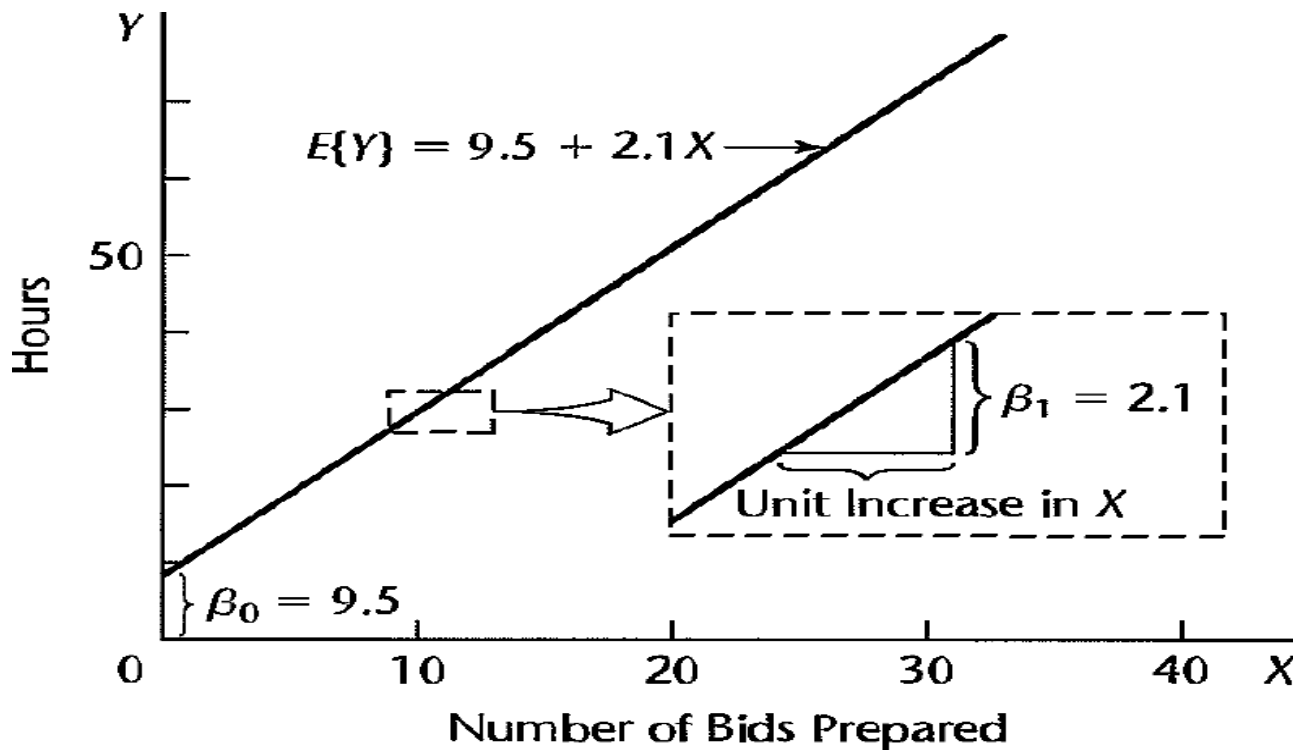
Example, cont'd

Suppose that in the i^{th} week, $X_i = 45$ bids are prepared and the actual number of hours required is $Y_i = 108$. In that case, the error term value is $\varepsilon_i = 4$, for we have $E\{Y_i\} = 9.5 + 2.1(45) = 104$ and $\varepsilon_i = 108 - 104 = 4$



Meaning of Regression Parameters

Regression coefficients: β_0 (intercept), β_1 (slope)

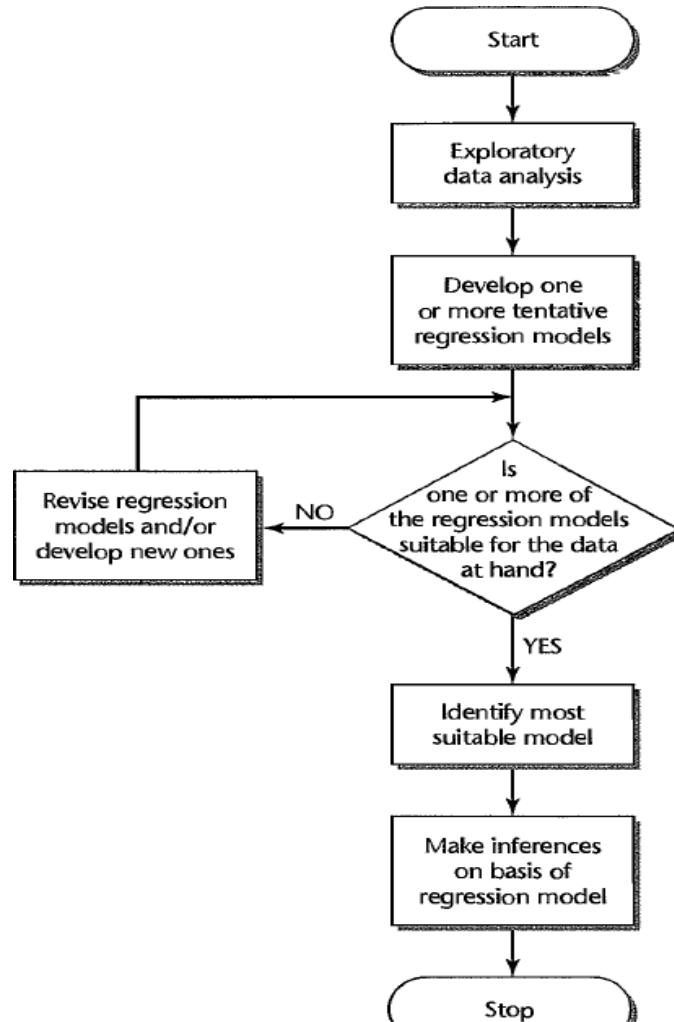


- β_0 gives the mean of the probability distribution of Y at $X = 0$. When the scope of the model does not cover $X = 0$, β_0 does not have any particular meaning as a separate term in the regression model.
- β_1 indicates that the preparation of one additional bid in a week leads to an increase in the mean of the probability distribution of Y of 2.1 hours.

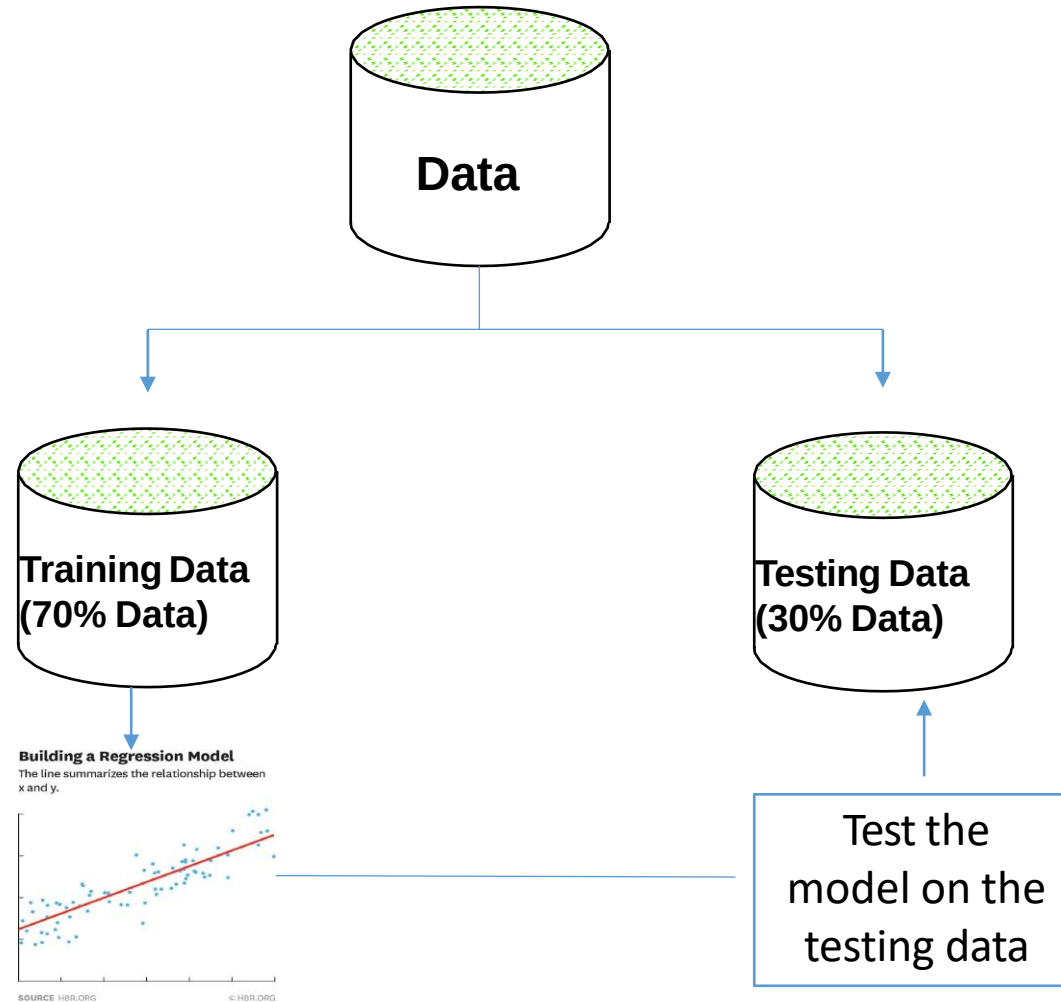
Data From Regression Analysis

- Unknown the regression parameters β_0, β_1
- Estimate parameters from relevant data
- Rely on an analysis of the data for developing a suitable regression model

Steps in Regression Analysis



Steps in Regression Analysis, cont'd

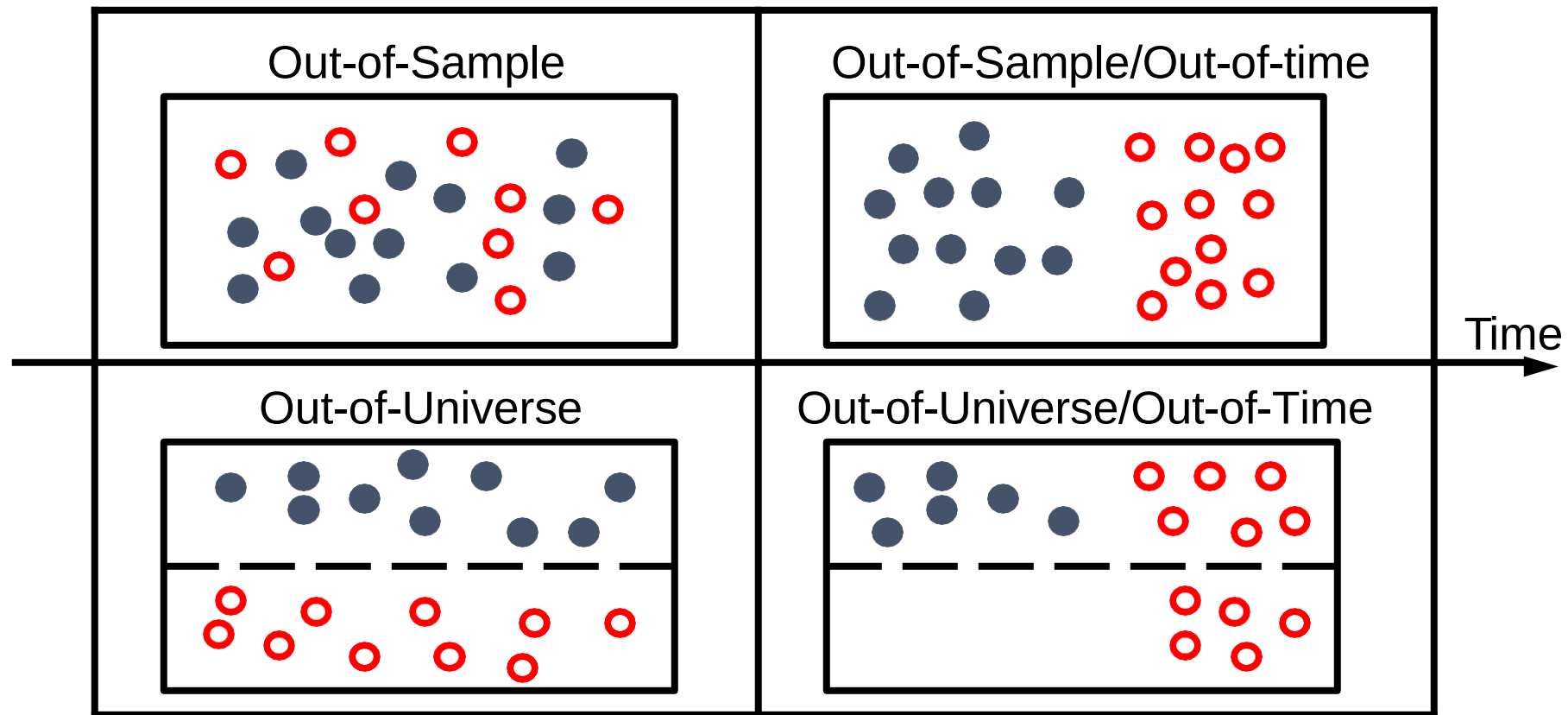


- Training Data is also called development sample.
- Testing Data is also called hold-out sample

Steps in Regression Analysis, cont'd

- **Out-of-Sample Tests**

● : Training set ○ : Test set



Estimate: Method of Least Squares

- For the observations (X_i, Y_i) for each case, the method of least squares considers the deviation of Y_i from its expected value:

$$Y_i - (\beta_0 + \beta_1 X_i)$$

- In particular, the method of least squares requires that we consider the sum of the n squared deviations. This criterion is denoted by Q :

$$Q = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

- According to the method of least squares, the estimators of β_0 and β_1 are those values b_0 and b_1 respectively, that minimize the criterion Q for the given sample observations $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$.

Example

In a small-scale study of persistence, an experimenter gave three subjects a very difficult task. Data on the age of the subject (X) and on the number of attempts to accomplish the task before giving up (Y) follow:

Subject i :	1	2	3
Age X_i :	20	55	30
Number of attempts Y_i :	5	12	10

In terms of the notation to be employed, there were $n = 3$ subjects in this study, the observations for the first subject were $(X_1, Y_1) = (20, 5)$, and similarly for the other subjects.

Example, *cont'd*

What is the number of attempts for 25 year old? How can you answer this question?

Method 1: Overall Average, $\bar{Y} = \frac{5+12+10}{3} = 9$

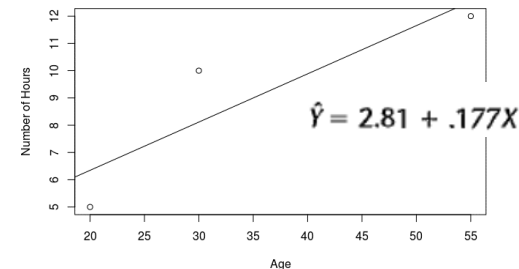
Answer is 9 attempts.

Method 2: Average of 20 and 30 years old, $\bar{Y} = \frac{5+10}{2} = 7.5$

Answer is 7.5 attempts.

Method 3: Use regression model

Answer is $2.81 + 0.177 \cdot 25 = 7.235$



Example, *cont'd*

What is the number of attempts for 25 year old? How can you answer this question?

Method 1	Method 2	Method 3
9	7.5	7.235

Which one would you choose?

Lets use these methods to predict Y for our data set. In other words, lets evaluate the in sample performance with actual values of X and Y.

Example, *cont'd*

What is the number of attempts for 25 year old? How can you answer this question?

Lets look at the in sample performance of all 3 methods

X	Y	Method1 Error	Method 2 Error	Method 3 Error	Min Error
20	5	$5-9 = -4$	$5-7.5=-2.5$	$5-(2.81+0.177*20)=5-6.35=-1.35$	Method 3
55	12	$12-9= 3$	$12-7.5 =4.5$	$12-(2.81+0.177*55)=12-12.53=-0.53$	Method 3
30	10	$10-9=1$	$10-7.5=2.5$	$10-(2.81+0.177*30)=30-8.115=1.88$	Method 1

Which one would you choose?

Is there an aggregated number which would summarize the model performance?

Example, *cont'd*

Which method would you choose?

Metric	Method 1 Error	Method 2 Error	Method 3 Error
Average	$(-4+3+1)/3=0$	$(-2.5+4.5+2.5)/3=1.5$	$(-1.35-0.53+1.88)/3=0$
Sum	$(-4+3+1) = 0$	$(-2.5+4.5+2.5)=4.5$	$(-1.35-0.53+1.88)=0$
Absolute (Sum)	$(4+3+1) = 8$	$(2.5+4.5+2.5)=9.5$	$(1.35+0.53+1.88)=3.76$
Squared Sum	$16+9+1=26$	32.75	5.65

Which metric would you choose?

- Easy to handle and nice mathematical properties (e.g. derivatives exists)
- Squared Sum of Residuals is the common choice for the regression models.

Least Squares Estimation

$$Q = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

- How to calculate β_0 and β_1 from the equation above?
- Need to calculate the values of β_0 and β_1 that minimize Q
- Our objective is to minimize Q !

Least Squares Estimation

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad i = 1, \dots, n$$

$$\Rightarrow \varepsilon_i = Y_i - (\beta_0 + \beta_1 X_i)$$

$\equiv$ function of unknown β_0, β_1 and observed data

$$Q = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

Goal: select values of β_0, β_1 that minimize Q and label them as b_0, b_1

$$(i): \frac{\partial Q}{\partial \beta_0} = 2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)(-1) \stackrel{\text{set}}{=} 0 \Rightarrow \sum_{i=1}^n Y_i = nb_0 + b_1 \sum_{i=1}^n X_i$$

$$(ii): \frac{\partial Q}{\partial \beta_1} = 2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)(-X_i) \stackrel{\text{set}}{=} 0 \Rightarrow \sum_{i=1}^n X_i Y_i = b_0 \sum_{i=1}^n X_i + b_1 \sum_{i=1}^n X_i^2$$

Least Squares Estimation , Cont'd

Solving (by multiplying (i) by $\sum_{i=1}^n X_i$ and (ii) by n and taking (ii) – (i)) :

$$n \sum_{i=1}^n X_i Y_i - \sum_{i=1}^n X_i \sum_{i=1}^n Y_i = b_1 \left(n \sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i \right)^2 \right) \Rightarrow$$

$$\sum_{i=1}^n X_i Y_i - \frac{\sum_{i=1}^n X_i \sum_{i=1}^n Y_i}{n} = b_1 \left(\sum_{i=1}^n X_i^2 - \frac{\left(\sum_{i=1}^n X_i \right)^2}{n} \right) \Rightarrow$$

$$b_1 = \frac{\sum_{i=1}^n X_i Y_i - \frac{\sum_{i=1}^n X_i \sum_{i=1}^n Y_i}{n}}{\sum_{i=1}^n X_i^2 - \frac{\left(\sum_{i=1}^n X_i \right)^2}{n}} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{SS_{XY}}{SS_{XX}} = \sum_{i=1}^n \frac{(X_i - \bar{X})}{SS_{XX}} Y_i = \sum_{i=1}^n k_i Y_i$$

$$\text{From (i): } b_0 = \bar{Y} - b_1 \bar{X} = \sum_{i=1}^n \left(\frac{1}{n} + \frac{\bar{X}(X_i - \bar{X})}{SS_{XX}} \right) Y_i = \sum_{i=1}^n l_i Y_i$$

Properties of Least Squares Estimation

- An important theorem, called the Gauss Markov theorem, states: Under the conditions of regression model, the least squares estimators b_0 and b_1 are unbiased and have minimum variance among all unbiased linear estimators.
 - **Unbiased:** $E\{b_0\} = \beta_0; \quad E\{b_1\} = \beta_1$
 - **More Precise:** The estimators b_0 and b_1 are more precise than any other estimators belonging to the class of **unbiased estimators** that are linear functions of the observations $Y_1, \dots, Y_n$. Linear function of Y_i :
 - $b_1 = \sum k_i Y_i$
 - $b_0 = \sum l_i Y_i$
 - $E(Y_i) = \sum (l_i + k_i) Y_i$
- Since the k_i and l_i are known constants (because the X_i are known constants), b_0 and b_1 are a linear combination of the Y_i and hence are linear estimators.

Point Estimation of a Mean Response

- **response**: a **value** of the response variable **Mean response**: $E\{Y\}$
- **Estimated regression function**:

$$\hat{Y} = b_0 + b_1 X$$

($\hat{Y}$: the value of the estimated regression function at X of the predictor variable and an unbiased estimator of $E\{Y\}$)

Fitted value $\hat{Y}_i$:

$$\hat{Y}_i = b_0 + b_1 X_i \quad i = 1, \dots, n$$

Fitted Values and Residuals

True Regression Function: $E \{Y\} = \beta_0 + \beta_1 X$ (Unknown, since $\beta_0, \beta_1 \equiv$ parameters)

Estimated Regression Function (Fitted): $\hat{Y} = b_0 + b_1 X$

For the i^{th} observation: $\hat{Y}_i = b_0 + b_1 X_i \quad i = 1, \dots, n$

Residuals: Differences between observed and fitted (predicted) values:

$$e_i = Y_i - \hat{Y}_i = Y_i - (b_0 + b_1 X_i) \quad i = 1, \dots, n$$

Properties of Residuals:

$$\sum_{i=1}^n e_i = 0 \quad (\text{From LS eq (i)})$$

$$\sum_{i=1}^n X_i e_i = 0 \quad (\text{From LS eq (ii)})$$

$$\Rightarrow \sum_{i=1}^n \hat{Y}_i e_i = \sum_{i=1}^n (b_0 + b_1 X_i) e_i = b_0 \sum_{i=1}^n e_i + b_1 \sum_{i=1}^n X_i e_i = 0$$

Estimating Error Variance σ^2

$$\sigma^2 = \sigma^2 \{\varepsilon\} = E \left\{ \left(\varepsilon - E \{ \varepsilon \} \right)^2 \right\} = E \left\{ (\varepsilon - 0)^2 \right\} = E \{ \varepsilon^2 \}$$

ε unobservable since $\varepsilon = Y - (\beta_0 + \beta_1 X)$

We use residual e to "estimate" ε

$$e = Y - \hat{Y} = Y - (b_0 + b_1 X)$$

Obtain the "average" squared residual to estimate σ^2 :

$$s^2 = \frac{\sum_{i=1}^n e_i^2}{n-2} = \frac{\sum_{i=1}^n \left(Y_i - \hat{Y}_i \right)^2}{n-2} = \frac{SSE}{n-2} = MSE$$

Summary

- We have not made any assumptions regarding the distribution of the data
- Our objective function is the sum of least squares errors.
- We calculate the intercept and slope by finding the values intercept and slope that minimize the sum of the least square of errors.

Normal Error Regression Model

The normal error regression model:

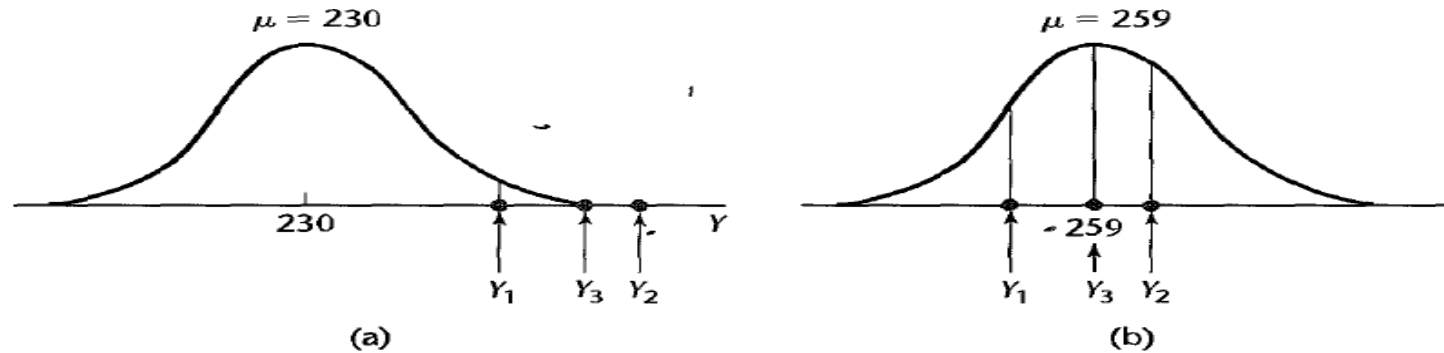
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Y_i : the observation response
- X_i : a known constant
- β_0, β_1 : parameters
- $\varepsilon_i, i = 1, \dots, n$: independent $N(0, \sigma^2)$

The estimators of the parameters β_0, β_1 and σ^2 can be estimated by the method of *maximum likelihood*. (MLE)

Normal Error Regression Model, cont'd

FIGURE 1.13
Densities for
Sample
Observations
for Two
Possible Values
of μ : $Y_1 = 250$,
 $Y_2 = 265$,
 $Y_3 = 259$.



The method of maximum likelihood chooses as the maximum likelihood estimate that value for which the likelihood value is largest.

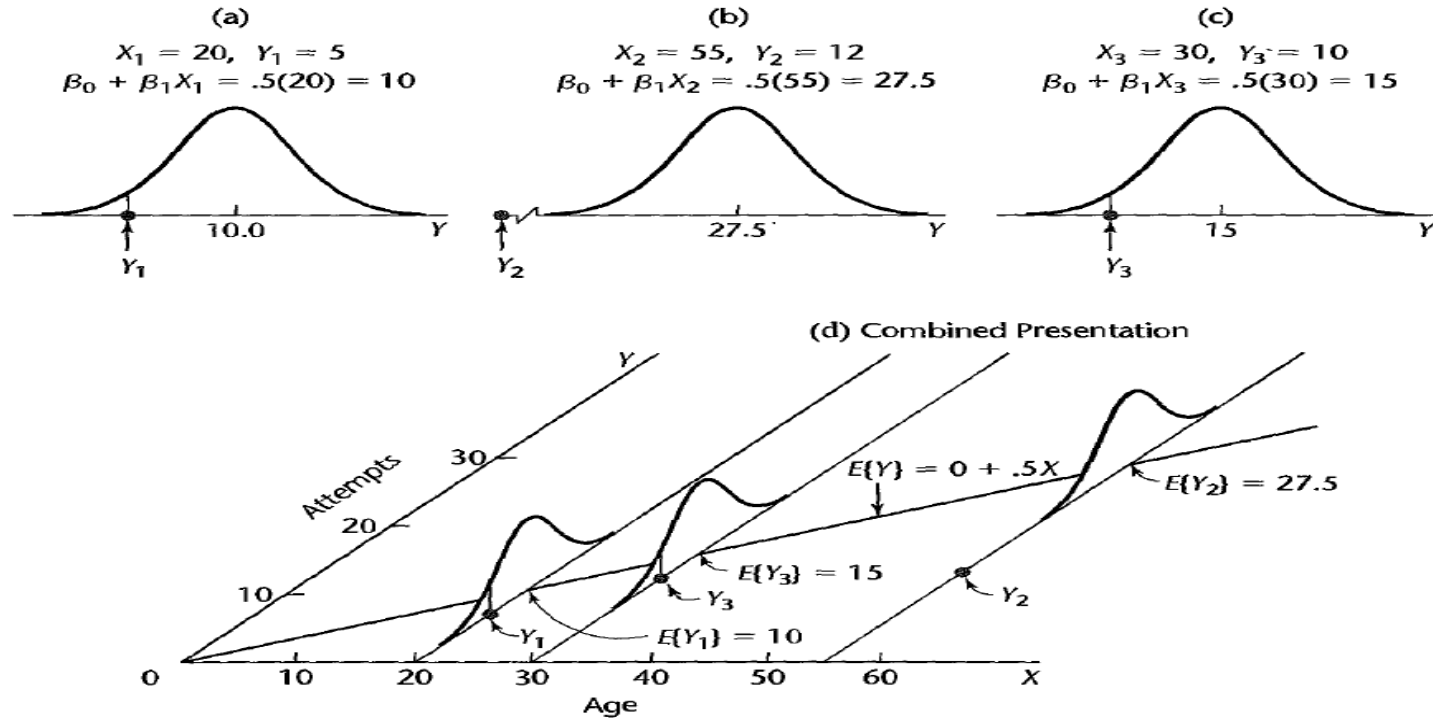
Two methods for finding MLE:

1. a systematic numerical search
2. use of an analytical solution

Estimator of μ is the sample mean $\bar{Y}$

Normal Error Regression Model, cont'd

FIGURE 1.15 Densities for Sample Observations if $\beta_0 = 0$ and $\beta_1 = 5$ —Persistence Study Example.



$$\sigma = 2.5; \beta_0 = 0; \beta_1 = 0.5$$

Normal Error Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad i = 1, \dots, n \quad \varepsilon_i \sim N(0, \sigma^2) \quad (\text{independent})$$

$$f(y_i) = f_i = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{1}{2} \left(\frac{y_i - (\beta_0 + \beta_1 X_i)}{\sigma} \right)^2 \right\} \quad i = 1, \dots, n$$

Likelihood Function $\Rightarrow L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n f_i = (2\pi)^{-n/2} \sigma^{-n} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n \left(\frac{y_i - (\beta_0 + \beta_1 X_i)}{\sigma} \right)^2 \right\}$

Goal: Choose values $\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2$ that maximize L (or equivalently $l = \ln(L)$):

Log Likelihood Function $\Rightarrow l = \left(-\frac{n}{2} \right) \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2} \sum_{i=1}^n \left(\frac{y_i - (\beta_0 + \beta_1 X_i)}{\sigma} \right)^2$

Note: maximizing l wrt β_0, β_1 is same as minimizing $\sum_{i=1}^n \left(\frac{y_i - (\beta_0 + \beta_1 X_i)}{\sigma} \right)^2$

$$\Rightarrow \hat{\beta}_0 = b_0, \quad \hat{\beta}_1 = b_1$$

$$\frac{\partial l}{\partial \sigma^2} = -\frac{n}{2} \left(\frac{1}{\sigma^2} \right) - (-1) \frac{1}{2} \sum_{i=1}^n \frac{(y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2}{(\sigma^2)^2} \stackrel{set}{=} 0 \Rightarrow \hat{\sigma}^2 = \frac{\sum_{i=1}^n \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i) \right)^2}{n} = \frac{\sum_{i=1}^n e_i^2}{n} = \frac{n-2}{n} s^2$$

Example: Toluca Company

The Toluca Company manufactures refrigeration equipment as well as many replacement parts. In the past, one of the replacement parts has been produced periodically in lots of varying sizes. When a cost improvement program was undertaken, company officials wished to determine the optimum lot size for producing this part. The production of this part involves setting up the production process (which must be done no matter what is the lot size) and machining and assembly operations. One key input for the model to ascertain the optimum lot size was the relationship between lot size and labor hours required to produce the lot. To determine this relationship, data on lot size and work hours for 25 recent production runs were utilized.

The production conditions were stable during the six-month period in which the 25 runs were made and were expected to continue to be the same during the next three years, the planning period for which the cost improvement program was being conducted.

R Studio Cloud - R Commands

Upload Toluca Data to R Studio Cloud

```
# attaching the dataframe
attach(toluca_data)
# fitting the regression model
toluca.reg <- lm(workhrs ~ lotsize)
# getting the summary regression output:
toluca.reg <- lm(workhrs ~ lotsize)
# getting the ANOVA table:
anova(toluca.reg)
# getting the fitted values:
fitted(toluca.reg)
# getting the residuals:
names(toluca.reg)
resid(toluca.reg)
toluca.reg$residuals
plot(lotsize, workhrs)
# overlaying the regression line on this scatter plot:
abline(toluca.reg)
```

Example: Antioxidant Data – One Factor Analysis

- Dataset: lager_antioxidant_reg.csv
- Source: H. Zhao, H. Li, G. Sun, B. Yang, M. Zhao (2013). "Assessment of Endogenous Antioxidative Compounds and Antioxidant Activities of Lager Beers," Journal of the Science of Food and Agriculture, Vol. 93, pp. 910-917.
- Description: Total phenolic content, melanoidin content, various measures of antioxidant activity in 40 lager beers.
- Variables/Labels
 1. Beer ID (beer)
 2. Total phenolic content (tpc)
 3. Melanoidin content (ma)
 4. DPPH radical scavenging activity (dsa)
 5. ABTS radical cation scavenging activity (asa)
 6. Oxygen radical absorbance activity (orac)
 7. Reducing Power (rp)
 8. Metal Chelating Activity (mca)
- Perform one factor analysis by finding the best variable to explain Total Phenolic Content (TPC). Fit one variable regression model with TPC as a dependent variable against remaining variables above, as an independent variable one at a time. For example, $TPC = b_0 + b_1 MA$, $TPC = b_0 + b_1 DSA$ and soon.

Code – Program in R (beer example)

```
prg1<-function(x){  
  out1<-list({})  
  out2<-data.frame(matrix(0,nrow=6,ncol=2))  
  for (i in 1:6){  
    out1[[i]]<-lm(x[,2]~x[,2+i],data=x)  
    out2[i,<cbind(dimnames(x)[[2]][2+i],round(sum(out1[[i]]$residuals^2),2))  
  }  
  dimnames(out2[[2]])[[2]]<c("X","SSE")  
  list(out1,out2)  
}  
prg1(lager_antioxidant_reg)
```